

Nil Crespo Peiró

Senior Computer Vision Engineer | Real-Time Tracking | Sports AI

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PROFILE

Senior Computer Vision Engineer with academic and production experience building real-time sports-tracking systems. Combines technical leadership with hands-on model development, video-based tracking, model deployment, quality evaluation, distributed processing, and analyst-facing products across basketball and padel.

EXPERIENCE

Senior Computer Vision Engineer

Catapult / IMPECT | Aug 2026 - Present | Barcelona, Spain (Remote)

- Lead technical direction of computer-vision and tracking systems for movement classification and performance analysis.
- Manage and mentor a small engineering team while remaining hands-on with system design and development.
- Train, optimize, and deploy real-time segmentation and classification models in production environments.
- Extend tracking systems to new movements, metrics, and analysis types.
- Evaluate tracking accuracy and build quality metrics to improve reliability across the system.

Researcher and PhD Candidate

Deutsche Sporthochschule Köln | Sep 2024 - Aug 2026 | Cologne, Germany

- Developed an automated pipeline for a large-scale 3x3 basketball player-tracking dataset from single-camera broadcast footage.
- Designed game-segment detection, court registration, player tracking, 3D ball localization, and court-coordinate projection methods.
- Reduced identity fragmentation across cuts and occlusions using re-identification, jersey recognition, appearance features, and spatiotemporal constraints.
- Produced a tracking dataset forming part of an upcoming publication.

Computer Vision Developer

AiBall | Jun 2023 - Jul 2026 | Barcelona, Spain

- Developed production player and ball tracking for padel, including multi-camera processing and 3D ball-trajectory reconstruction.
- Designed backend models and APIs for clubs, courts, cameras, reservations, players, and matches.
- Integrated reservation platforms and automated match creation, recording, and processing workflows.
- Built operational dashboards and monitoring for data quality, processing, camera/NVR health, and club performance.
- Contributed to distributed video processing with cloud storage, background workers, and automated alerts.

Computer Vision Intern

Arcvi | Dec 2022 - Jul 2023 | Barcelona, Spain

Data Analyst

BallandStats | Oct 2022 - Dec 2023

- Built FEB game-data scraping and update workflows, a relational schema, and team-specific visualizations.

Lead Data Analyst

Barcelona Dragons | Apr 2022 - Sep 2023

- Created an advanced-statistics warehouse and pre-game KPI analysis with coaching staff.

Data Scientist

HP | Sep 2021 - Apr 2023

- Structured complex data flows and applied predictive models to unstructured business data.

Artificial Intelligence Intern

Basetis | Mar 2021 - Jul 2021

- Worked on ranking, recommendation, and NLP similarity algorithms and delivered a supporting web application.

SELECTED PROJECTS

Real-Time Athlete Computer Vision

Catapult / IMPECT | Ongoing

- Real-time segmentation, classification, athlete tracking, quality evaluation, and performance analysis.

3x3 Basketball Tracking from Broadcast Video

Deutsche Sporthochschule Köln | 2024 - 2026

- Automated game segmentation, court registration, player tracking, 3D ball localization, and dataset development.

Production Padel Tracking Platform

AiBall | 2023 - 2026

- Multi-camera tracking, 3D ball reconstruction, distributed processing, monitoring, and automated club workflows.

Team Canada Olympic Rotation Analytics

Canadian Men's Basketball Team | 2024

- Rotation, lineup, and game-dynamics visualizations supporting Olympic preparation and review.

BUEStats 2.0 - Spanish Basketball Intelligence

Personal project with Marc Cortadellas | Launched

- FEB data collection, scouting views, shot charts, rankings, comparisons, and player-similarity tools.

NBA Player Value and Prediction Lab

Personal project | 2024

- Interactive studies of salary efficiency, future-star prediction, and league scoring trends.

EDUCATION

Master's Degree - Sports Analytics

UCAM Universidad Católica San Antonio de Murcia | Sep 2022 - Jun 2023

- Artificial Intelligence applied to Sports.

Erasmus - Mathematics and Computer Science

Technische Universität Darmstadt | Sep 2020 - Mar 2021

Bachelor - Data Science and Engineering

Universitat Politècnica de Catalunya | 2017 - 2021

TECHNICAL SKILLS

Computer vision: segmentation, classification, player and ball tracking, re-identification, jersey-number recognition, 3D trajectory reconstruction, camera calibration, and movement classification.

ML and production: Python, scikit-learn, Keras, NumPy, Pandas, SciPy, SQL, efficient real-time inference, model optimization, evaluation, deployment, AWS, cloud storage, and background workers.

Data products: Streamlit, Dash, Power BI, Tableau, scraping, relational modeling, monitoring, dashboarding, data warehousing, HTML, CSS, and JavaScript.

LANGUAGES

Catalan (Native/Bilingual) | Spanish (Native/Bilingual) | English (C1) | German (B2)